



Federated meta heuristic cryptography algorithm with fuzzy logic model in cloud secure data migration

Kannan Srinivasan¹ · Guman Singh Chauhan² · Rahul Jadon³ · Rajababu Budda⁴ · Venkata Surya Teja Gollapalli⁵ · Joseph Bamidele Awotunde⁶

Received: 30 October 2024 / Revised: 24 June 2025 / Accepted: 2 July 2025 / Published online: 9 October 2025
© The Author(s), under exclusive licence to Springer Science+Business Media, LLC, part of Springer Nature 2025

Abstract

In the modern world, cloud computing has profoundly changed almost every aspect of our lives and organizational structures. It is without a doubt one of the strategic trajectories for many businesses as well as most prevalent infrastructure for businesses as well as end users. One of the most well-liked cloud computing services is data outsourcing, which is continuously done to public or hybrid clouds. It's essential to provide the necessary safeguards and handle cloud data carefully. Security must be implemented at every stage of data transfer process, including data discovery, classification, cataloguing of mission-critical data access. Therefore, this research proposed model develops novel technique in secure data migration in cloud network based on machine learning (ML) with cryptography methods. Here data migration is carried out in secure cloud environment using federated adversarial reinforcement cryptography algorithm with data optimization by fuzzy logic based firefly whale meta-heuristic binary swarm algorithm. the experimental analysis has been carried out based on various cloud network security analysis dataset in terms of packet delivery ratio, latency, throughput, QoS, data integrity. Effectiveness of the suggested framework when compared to current methods yields robust outcomes and demonstrates its increased security. The proposed model achieved 98% throughput, 96% packet delivery ratio, 97% latency, 95% quality of service, and 98% data integrity. The existing **PBE-ENCR** attained Packet delivery ratio (PDR) of 84%, Data integrity of 81%, Throughput of 86%, LATENCY of 85%, QOS of 87%, **SVM-GA** PDR of 90%, Data integrity of 93%, Throughput of 91%, LATENCY of 88%, QOS of 92%.

Keywords Secure data migration · Cloud network · Machine learning · Cryptography techniques · Adversarial reinforcement

✉ Kannan Srinivasan
kannansrinivasan@ieee.org

Guman Singh Chauhan
gumansinghchauhan@ieee.org

Rahul Jadon
rahuljadon@ieee.org

Rajababu Budda
rajababubudda@ieee.org

Venkata Surya Teja Gollapalli
venkatasuryatejagollapalli@ieee.org

Joseph Bamidele Awotunde
jbawotunde@gmail.com

¹ Saiana Technologies Inc, Piscataway, NJ, USA

² John Tesla Inc, Dallas, TX, USA

³ CarGurus Inc, Boston, MA, USA

⁴ IBM, San Francisco, CA, USA

⁵ Centene Management Company LLC, Clayton, MO, USA

⁶ Department of Computer Science, Faculty of Information and Communication Sciences, University of Ilorin, Ilorin 240003, Kwara State, Nigeria

1 Introduction

In recent years, there are significant evolution in use of distributed computing systems and technology. This development has led to the creation of numerous distributed network methods, architectures, infrastructures, including network, Pervasive, Autonomic, cloud, etc. Cloud computing is the term for a computing network that is usually connected by the Internet and shares a decentralized quantity of services provided by a to meet demands of user [1]. NIST definition of cloud computing refers to a framework that offers ubiquitous, quick, and on-demand network access to a shared set of reconfigurable computing resources (such software). Scalability, affordability, and manageability are some advantages of cloud computing. Cloud storage also offers infrastructural features including ease of use, economics, universality, lease pluralism, dependability, and versatility. These technologies can be used by a cloud client upon request to create, execute, and host services and apps that are adaptable to any location or device. NIST “concept emphasizes the three service models – Infrastructure-as-a-Service (IaaS), Platform-as-a-Service (PaaS), Software-as-a-Service (SaaS) which presenting cloud services to the client through a Cloud Service Provider (CSP) customers. It also briefs the four deployment models — Public Cloud and Hybrid Cloud, Private Cloud, Community Cloud, emphasizing the paradigm for distributing computing assets to coordinate the cloud administration [2]. On the other hand, cloud infrastructure faces a range of problems primarily related to interoperability, scalability, and multi-tenancy. But the most critical issues apply to protection as cloud infrastructure is exposed to a variety of threats as a system utilizing internet networks (such as embedded networks, grid computing, etc.)” [3]. Security issues with cloud computing will prevent it from being widely adopted. The sharing of cloud computing services actually makes it more difficult to maintain their security and prevent unauthorized use or access. This problem primarily affects data that is outsourced to the cloud. One of the main security concerns with cloud computing is network security, which has to do with both internal and external assaults. Sharing or transferring data from one end to another is backbone of present information technology era Cloud computing is a network based technology based upon pay per usage model and most enticing technology these days. Broad network access, storage capacity, processing power, user and application resources, etc. are all made possible by cloud computing [4]. Due to quick growth in technologies more and more service providers and customers migrating towards cloud environment. As security is one of the important aspects that are considered in network based technologies, so same is the concern that is the hindrance in the wide acceptance of cloud computing. Securely

transferring of data need a mechanism to encrypt or ciphering data. Data which is stored in cloud is not much secure according to the user’s point of view. Cryptography is considered as one of the solutions for the data security. Reliability of cloud data is contingent upon the use of trusted public keys. A trustworthy public access key is produced utilizing a variety of techniques, including TLS (transport layer security) and SSL (secure socket layer). A well-developed peer-to-peer web communication standard is TLS. A hostile attack that exploits holes in TLS usage can lower the reliability of TLS. The public-key infrastructure (PKI) that TLS uses is one example of such a situation. Any website can be impersonated by a single reliable party. Traditionally, a public key (TLS PKI) is validated by a trusted certification authority (CA). A CA must confirm that an identity and a public key are bound in order to issue an authentication certificate. Regretfully, this approach does not offer a high level of security confidence for a variety of reasons. Using the trust on the first usage approach, identity verification is done online [5]. This verification happens once for every certificate (i.e., each certificate issuance). Consequently, an attacker with a brief window of opportunity to assume the identity of a domain can get a domain authentication certificate. Although the cloud is more flexible and agile than traditional computer systems, its introduction of new security threats makes consumers worry about security and dependability. Service models, end service features, and deployment patterns all affect the security of this computer architecture. Comparing the cloud platform’s security and privacy to traditional information security data systems, the former is more crucial. Your company’s use of cloud services including data management systems, automatic software upgrades, quick data access to critical information, and user-based growth services may be being pushed by cloud computing data analytics. Our clients have reaped incredible benefits as a consequence. Cloud computing is having an impact on a number of industries, including e-commerce, e-learning, and healthcare. offers a quick and affordable internet connection. This might be the next big technological and industrial revolution. Cloud computing is being used by an increasing number of e-commerce businesses to provide convenience [6].

Research contribution Our primary objective is on finding solutions for challenges related to the design and development of scalable architectures, as well as the understanding of application migration and cost-effective and efficient migration methods. Aim of this research is to provide a novel technique based on machine learning and cryptography for secure data movement in cloud networks. Here, the data movement is done in a secure cloud environment utilising the firefly whale meta-heuristic binary swarm technique

based on fuzzy logic for data optimisation and federated adversarial reinforcement cryptography. Firefly Whale Meta-Heuristic algorithm is a firefly and humpback whale-inspired optimization algorithm. The firefly algorithm assists in exploring the potential solutions, while the whale algorithm optimizes them based on a bubble-net hunting mechanism to enhance the efficiency and performance of data migration. It is also possible to modify the number of rotations and the length of the secret key. The experimental results of the proposed algorithm showed a high degree of security and an apparent improvement in measurements of cipher execution time and security forces when compared to the cryptographic techniques typically employed in cloud computing.

2 Background and related works

Data security and standardization are two crucial areas for sustainability in the current cloud ecosystem. Considering the complexity of the architecture of cloud computing, they are still in their early stages and offer a lot of potential for future research. Security of transit data from corporate to cloud storage identified as the area of concern, according to study studies on cloud security and related topics. Many studies have been conducted on how to safeguard data while it is at rest, whether in an enterprise setting or on the cloud, but organizations are still concerned about the security of data while it is being transmitted. Although particular requirements can be met with approaches for safe data transfer, there is currently no framework in place to handle all forms of data migration from a private setting to the public cloud [7]. The approach proposed in [8] outlines a method for improve that protection that of user information in relation to cloud computing by utilizing encryption methods, including diverse algorithms. The authors recommended a set that of various algorithms to deal with the problems regarding confidentiality, isolation, and loss of information. A Lightweight Feistel Structure with Substitution Permutation has been introduced by Work [9] to apply security measures in the cloud. They have designed an undefined cryptographic architecture to safeguard multimedia data. By using a 64-bit block cypher with an identical key size, this paradigm seeks to maximise security while minimising computing overhead. The author of [10] pointed out that due of the cloud's improved software capabilities, consistent availability, increased flexibility, and lower prices, many organisations are turning to cloud solutions for data storage and management in this day of information technology. They emphasised how health care service systems are focussing more on patient-centeredness and modifying digital platforms. A modified form of AES that makes use of

MAC addresses is suggested in [11] for cloud data transport. Because MAC Address varies from machine to computer, it increases the randomness of AES processes. Ultimately, it is determined that even with knowledge of the primary key, the attacker was unable to easily decrypt the encrypted message. In [12], symmetric and asymmetric encryption were used, respectively, to carry out the processes of moving the files to and from the cloud. By employing the AES encryption technique, the RSA encryption reduces amount of time it takes to transmit data while also making it more difficult for hackers to crack. Even though cloud data security can be ensured by combining symmetric and asymmetric encryption algorithms, system overhead is caused by the need to double-encrypt and decrypt each file. The research in [13] identifies the several security flaws that currently exist in cloud computing environments and suggests a fresh approach to cloud data protection. The fundamental information security features, such as data confidentiality, authenticity, and access control, are suggested to be provided by the 128 bit AES encryption. Even though the suggested approach offers cloud data security, it has been noted that there is a marked rise in latency as file size increases. In [14], the author suggests using AES to improve security and protect the privacy of data stored in the cloud. It is also found that the amount and format of the files to be transferred affect how strong the AES encryption technique is. Initially, 128 bits of key encryption is advised for files with big data sizes, 192 bits for files with medium data sizes, and 256 bits for files with tiny data sizes. illustrating the range of data sizes after 14 iterations of encryption. Ultimately, the conclusion is that data format should determine the encryption strategy, rather than data size, when using a 256-bit key. Author [15] proposes a client-side encryption and decryption method that uses a single secret key to offer privacy and security for data that is stored on the cloud and that is to be sent. Ultimately, it was determined through the analysis of multiple approaches that the AES encryption and decryption mechanism ensures cloud security and privacy. Research on cloud-based IoT sensor nodes was published in Work [16], which also offered lack-IoT as a way to achieve direct device-to-device communication without the need for middlemen. It was discovered that the dependable access control technique cut down on communication and calculation time. The author [17] suggested utilising genetic algorithms to optimise energy consumption when migrating Internet of Things data to a cloud system. The suggested approach used less energy as compared to state-of-the-art algorithms. In a similar vein, research conducted by work [18] provided a thorough analysis of several threats that promote confidence in cloud computing and Internet of Things platforms. Positive theoretical analysis of the suggested trust model was conducted. An optimization platform [19]

integrating Ant Colony Optimization, Gradient Descent, and Bayesian Decision Models enhances resource allocation, run time, and scalability for cloud computing. Such methods are meta-heuristic extendable with methods such as the firefly-whale swarm in secure cloud data migration, maximizing efficiency and flexibility in distributed systems. Author [20] looked into the problems with sensor-cloud systems data privacy. This study demonstrated how bandwidth and storage costs can be decreased by privacy-based procedures linked to communication complexity. Work [21] discovered that while these methods produce good results, their high temporal complexity renders them less appropriate for use on computers in real environments. Differential privacy-based federated learning [22] supports secure decentralized training of models using noisy gradient updates, which prevents data reconstruction attacks. The method preserves privacy and accuracy levels with a low communication overhead, enabling secure management of data in distributed systems, and satisfies requirements for secure cloud migration of data in the proposed Federated Adversarial Reinforcement Cryptography Algorithm (FARCA) system. Using the bat algorithm [23], conducted their investigation. A hybrid technique is needed for load balancing as well as virtual machine optimisation. The goal of developing and implementing this swarm-based method is to balance load on physical computer by optimising load on virtual machine. Elliptic Curve Cryptography (ECC) offers [24] enhanced security with minimal key sizes and faster processing, addressing the issue of cloud computing being resource-constrained and inefficient. Its superior comparison to AES renders ECC most appropriate cryptography to ensure secure data transfer in cloud configurations, supporting hybrid cryptographic solutions like the federated adversarial approach in the uploaded study. An ABC method as well as heuristic method are combined in [25] implementation of a hybrid approach, which is regarded as a hybrid approach. It is crucial to take into account the amount of time needed to grow and distribute the load. Traditional cryptography methods experience overhead and scalability issues during cloud migration, whereas optimization algorithms hardly incorporate security. Federated learning improves privacy but is not adaptive encryption. The technique proposed here combines federated adversarial reinforcement learning with fuzzy logic metaheuristics to achieve adaptive encryption and data migration optimization. While this improves security, latency, and data integrity, the model can be linked with computational overhead, and its performance can degrade with varying cloud architectures, impacting scalability and flexibility. It is feasible to show that this method works well in situations that are homogeneous or heterogeneous.

Problem definition: For businesses moving from on-site infrastructure to cloud-based environments, the exponential growth of data poses significant challenges. Data migration, an essential step in the cloud adoption process, may present difficulties that keep the full benefits of the cloud from being exploited. These challenges include heterogeneity in the data, data security concerns, downtime minimisation, and cost optimisation. Problems associated with data transfer might be of many different kinds. Data heterogeneity—defined as data dispersed across multiple on-premise systems in various formats and architectures—represents a significant obstacle. Legacy systems may make use of proprietary structures or outdated data formats, which calls for complex data transformation and mapping to ensure compatibility with the desired cloud environment. This unpredictability means that careful identification, categorisation, and transformation of data is required to ensure seamless integration and functionality in the cloud. Data quality issues, such as redundancy and inconsistencies, can also significantly impact cloud data integrity and the migration process. Duplicate entries, outdated information, and missing values might diminish trust in the relocated data and complicate data analysis. Strict quality control and cleansing methods are required to ensure the completeness and accuracy of the migrated data. Data migration presents significant security challenges since companies need to ensure that sensitive data is safeguarded during the transfer using robust encryption and access control technologies. Given the possibility that conventional on-premise security perimeters won't translate seamlessly to the cloud, data security protocols might need to be reviewed. Strong encryption techniques must be used by businesses to protect data in the cloud, both in transit and at rest. Clear access control processes are also essential to ensure that, after transfer, only approved persons can access sensitive data.

3 Proposed secure data migration in cloud network

If an organisation wants to modernise its mission-critical applications as well as plans to migrate to cloud as part of that process, it wants to overcome the security challenges and avoid making the same mistakes as others. Severe security threats such as data leakage and man-in-the-middle attacks are countered by FARCA's use of federated adversarial reinforcement learning, which dynamically adapts encryption to prevent exposure of raw data during migration. While this approach effectively reduces risk, a clearer, more detailed description of specific threat scenarios would further highlight the specific security advantages FARCA offers. In order to optimise the chances of a successful cloud

migration, leverage these insights to address and consider a 10-point checklist of critical areas. While many programs fulfil utility needs and are undoubtedly non-core, there are also plenty that are integrated as well as support key business operations and services. The latter group might be prime candidates for immediate redeployment. Refactoring, often known as re-architecting or redesigning, is more appropriate for the previously integrated core ones. Our migration patterns consist of a series of architectural modifications made to the application deployment environment, which gradually modernise the existing application. We propose a template-based definition of migration patterns to enable clear pattern descriptions and to underpin pattern-based migration planning. Semantics of architectural methods both before and after migration served as the foundation for this definition. It could only be feasible to deploy application components in public clouds for specific migration patterns. Nonetheless, hybrid public/private platforms can be used to deliver the application for those patterns that take re-architecting into consideration.

4 Migration plan

The corporation is in charge of managing the current servers, networks, and related systems including cooling and power supply. In order to arrive at a manageable migration plan by implementing the suggested patterns, we provide a set of migration decisions and actions.

Step 1 Transfer the program, unaltered, to a cloud platform that offers dependable and available infrastructure. Although OS licenses and management expenses for maintaining hosted operating method taken into account, development costs are decreased because apps do not need to be rewritten. The migration patterns MP1, MP3, and MP4 were considered, and MP1 was chosen since it just requires a copy-as-is to cloud, negating the requirement for environmental services.

Step 2 Another option is to modify Expense such that it is hosted on a platform by a third party. This would save management expenses and the cost of moving the program to a new platform. Refactoring program to run in roles hosted in the cloud requires work. You can choose from MP5 to MP11. given that the user profiles had to be maintained on-site. Since there was no requirement for interface customisation or multi-cloud deployment, pattern MP6 was chosen.

Step 3 Give up on your custom payment app and hire a more general cloud service, which should be assessed for usability, security, and performance. While MP12, MP13,

and MP14 are suitable, MP13 is preferred because it requires integrating expenses with payment services.

Step 4 Cloud platforms' data storage capabilities are necessary for an external hosting decision. To store receipt images, Expense needs NoSQL storage and a relational database system. Because Azure SQL and Storage services satisfy the criteria, MP12 was chosen.

Step 5 For data access and monitoring, remote apps must be connected with on-premises systems and other cloud services. It may be necessary to integrate remote services in order to monitor a system using an authentication or systems operation tool. You can choose from MP7, MP8, MP9, MP12, MP13, MP14. A few modifications were required, therefore MP14 was chosen.

Step 6 Payment sub-system that is utilised by other programs must always be accessible, even when it is only used by employees for Expense. You can choose between MP10 and MP11, however if you want to prevent failover rebinding from developing, you can use a broker like in MP11.

Step 7 Utilising cloud-based value-added services like caching can optimise performance during data retrieval or store output, session state, and profile data in a cache. To support these cloud provider environmental services, MP3 was used.

Migration path. The result is the architecture in Fig. 1. A company may choose to combine multiple hosting, data storage, and cloud service options based on its needs. For instance, MP1 phase 1 takes the hosting strategy and gradually migrates to an Azure SQL Database by first using SQL Server hosted in a virtual machine. Instead of using relational databases, MP3 would take advantage of caching and table/blob storage capabilities to increase speed early rather than late.

5 Cloud environment security analysis using federated adversarial reinforcement cryptography algorithm (FARCA):

Federated adversarial reinforcement cryptography algorithm (FARCA) FARCA is an amalgamation of federated learning and adversarial reinforcement learning for safe data migration. It enables multiple parties to cooperate on a model without exposing raw data, utilizing reinforcement

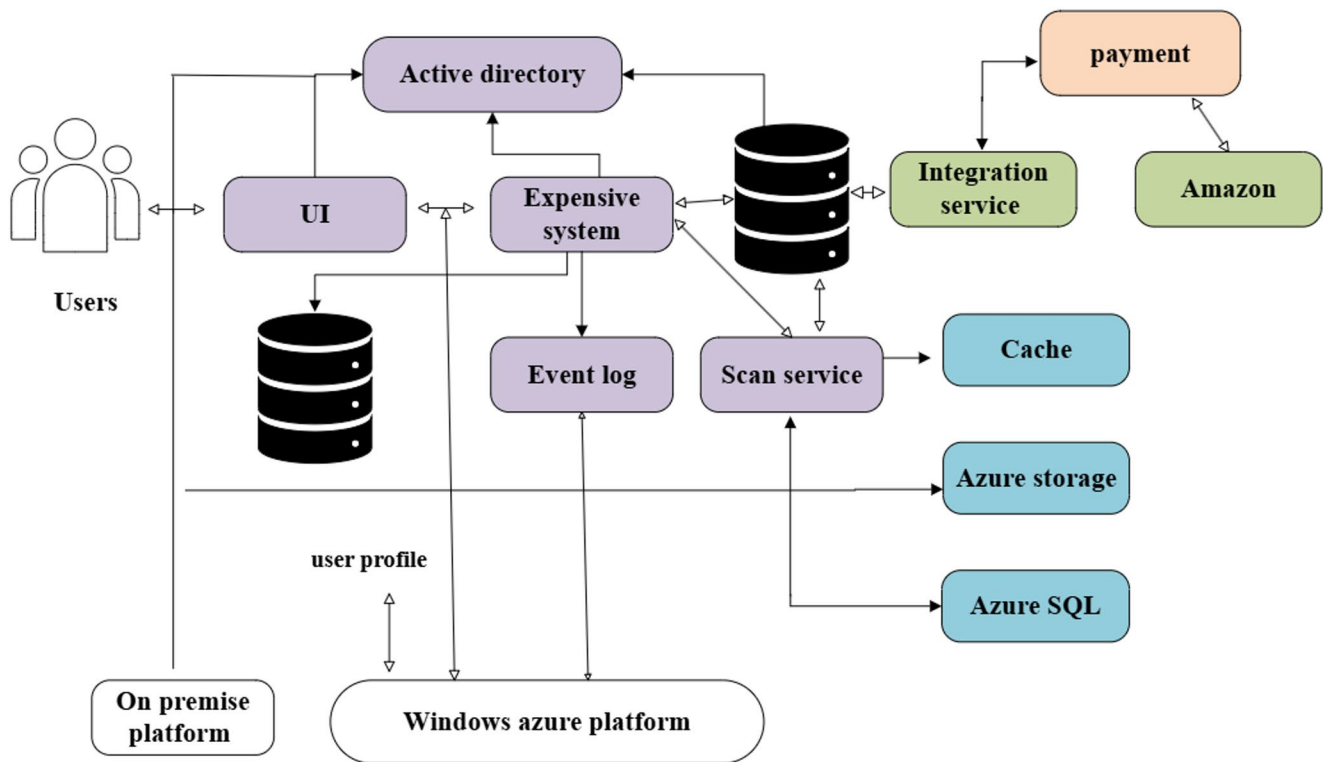


Fig. 1 Cloud network secure data migration architecture

learning to adjust encryption mechanisms for greater security over time.

The four entities that make up the system model are as follows: Data Owners (DOid), Cloud Platform (CP), Request Users (RUid), Third Party (T Pid).

1) DOid: The organisation that creates the data and contacts CP for services. Before uploading the data to CP, DOid encrypts it. DOid is regarded as an untrusted entity since it is thought that while it cannot leak its own data, it may leak data of another owner.

2) CP: An organisation that gathers all of the encrypted data from DOid and provides DOid or RUid with computing, sharing, and storage resources. For safe sharing between DOid or RUid, CP converts ciphertexts sent by DOid, does certain calculations over it, encrypts result. CP uses machine learning methods to train the collected data. Because it adheres to the protocol religiously and is genuinely interested in learning more, CP is seen as a semitrusted but untrusted entity in method. Two clouds comprise CP in our system paradigm: cloud1 is composed of Cloud Storage (CS) and Cloud Service Provider (CSP), while cloud2 houses the Classifier (CF).

3) RUid: An organisation that receives key and encrypted data from CP. It decrypts data it got from CP in order to extract the useful information. As an untrusted entity, RUid is regarded in the system model.

4) T Pid: An entity that is not authorised and untrusted and is connected to the system indirectly. T Pid can obtain the necessary data by either stealing the dataset from the legitimate source or by using a malicious entity.

Within our FedAN (federated adversarial network) paradigm, we present a partial sharing technique to mitigate the privacy issues associated with full GAN sharing. Specifically, we recommend training two generators: generator G_s for each user on the server side and generator G_u for local users. In order to bridge the training of two generators—one at a local user and the other at the server connected by a communication link—we use a common discriminator D_u , which is trained exclusively at the local user. In Fig. 2, we show user-server communication flow that results in a single-step update of the global model (CI), in order to graphically represent the FedAN updating process.

Procedure for user-server communication up to a single global classifier (CI) update utilizing the FedAN approach is displayed for a single user: First, a local cGAN is trained by local user. Following a single batch of training at local user, M_p is used to share trained D_u with the server. Where M is a publishing mechanism, $M_p : M(D_u, z, I)$. G_u is being trained at the current phase using the noise vectors z and the related fake labels I . We start a different generator G_s at the server for every user. We update G_s by one step for each user update that we get. Following N of these stages, we use artificial data produced from every G_s and data that

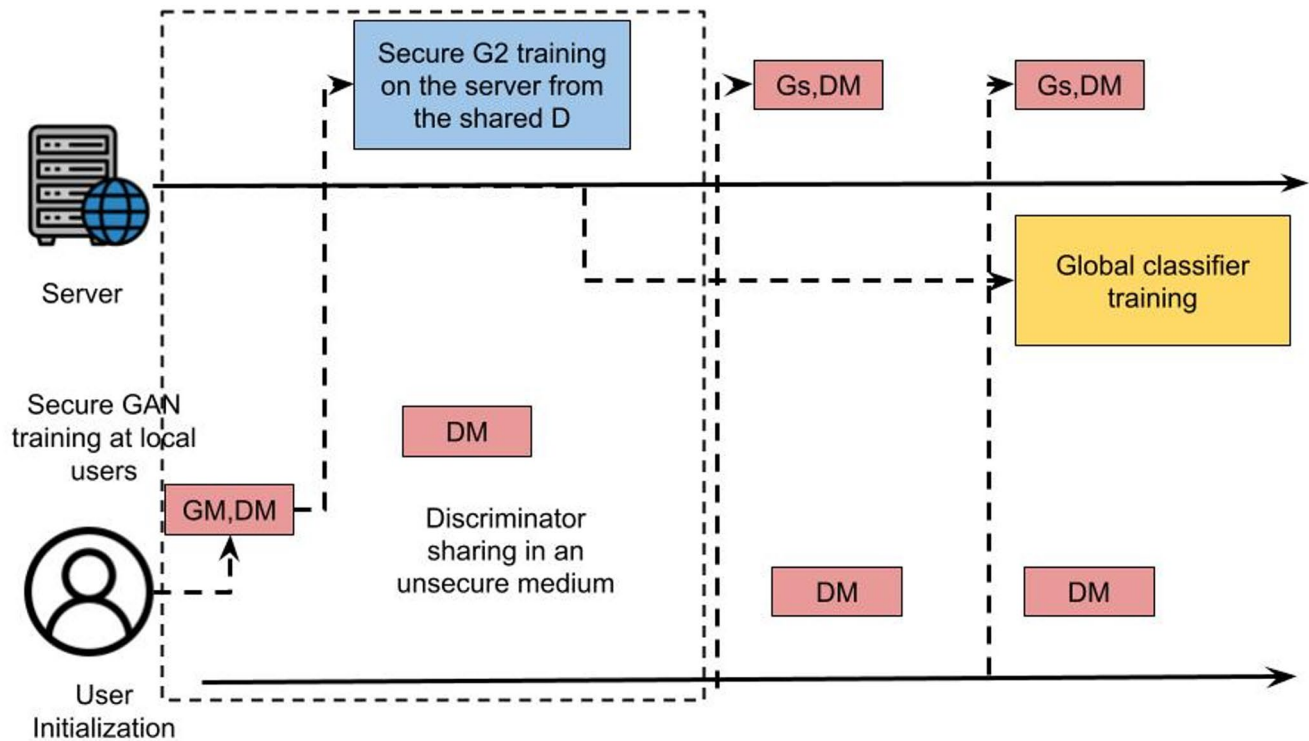


Fig. 2 User-server communication process

is available on the cloud to update the CI. This technique is carried out until CI converges.

When there is only one user, training process begins with local user teaching a local GAN method consisting of generator G_u as well as discriminator D_u . We base our study on the assumption that there is an unstable communication channel via which attackers might possibly gain access to the PS-FedAN publishing mechanism M_p . In accordance with FedAN standards, we initialise a unique generator G_s for every user on server side. The server then uses the data obtained from M_p to update appropriate G_s . We wait for a certain amount of changes from the user side, N , before updating global model CI. For the local user, N may denote an epoch in this context.

As a result, we simultaneously perform N updates on associated G_s and the local user's GAN. To update the global model CI, produced data from all updated G_s methods—which belong to various local users—is integrated with data that is available on server. Until CI converges to the intended point, this cycle of updates as well as combing through the generated data is repeated. We assume that the server and every local user have decided on the same architecture for G_u and G_s , as well as a secret seed, before installing the FL models. The weights of G_u and its matching G_s are started by the secret seed. Keep in mind that every user on the cloud server has a dedicated generator assigned to them. We also don't put any restrictions on D_u . FedAN uses

a batch-wise training procedure in which we continuously update G_s , G_u , D_u . D_u starts local GAN training after G_u and G_s are commenced. The FedAN publishing mechanism M_p is explained in detail in the ensuing subsections, which are followed by a description of the training procedure that is carried out for both the cloud server and the local users during each communication round.

Algorithm for Security analysis using federated adversarial network:

```

for every user  $i$  do
  Start  $G_{u_i}$  and  $G_{s_i}$  utilizing secret seed key  $y_i$ 
  Start discriminator:  $D_{u_i}$ 
end for
Start global model  $C_l$  (classifier) and train utilizing available cloud data.
for every communication round, until  $C_l$  is converged do
  for number of training iterations do
    for each step  $t$  do
       $z_t, l_t \leftarrow$  random vectors, random labels
      Train  $D_{u_i} : D_{u_i}^{(t+1)} \leftarrow \mathcal{F} \left( G_{u_i}^{(t)}(z_t, l_t), \mathcal{D}_{u_i}^{(t)} \right)$ ,
        where  $\mathcal{F}$  represent forward/back
        prop and weight updating
      Share  $\left( \mathcal{D}_{u_i}^{(t+1)}, z_t, l_t \right) : \mathcal{M}_p \left( \mathcal{D}_{u_i}^{(t+1)}, z_t, l_t \right)$ 
      Train  $G_{u_i} : G_{u_i}^{(t+1)} \leftarrow \mathcal{F} \left( G_{u_i}^{(t)}(z_t, l_t), D_{u_i}^{(t+1)} \right)$ 
      Train  $G_{s_i} : G_{s_i}^{(t+1)} \leftarrow \mathcal{F} \left( G_{s_i}^{(t)}(z_t, l_t), D_{u_i}^{(t+1)} \right)$ 
    end for
  end for
  Train  $C_l$  with cloud data and synthetic data  $C_l^{(t+1)} \leftarrow \mathcal{F} \left( C_l^{(t)} \right)$ 
end for

```

A quintuple $= (S, A, P, R, \gamma)$ indicate an MDP. Among these, S denotes the environment's state set, A the action set the agent is capable of doing, P set of state-action pairs that can be executed. A reward function, denoted by R , represents expected reward value earned while completing action a in state s and subsequently transferring to state s' . A transfer function, denoted by P , reflects probability of moving to state s' after performing action a in state s . The particular interaction process between the agent and the environment is split into two sections: 1) At time t , the agents observe state of external environment as it is at $s_t \in A$. It then chooses, in accordance with the strategy π , the action, $a_t \in A$, to carry out. 2) After updating the state s' in response to agent, environment moves on to next MDP process and produces a reward or return, denoted as $r_t \in \mathbb{R}$.

In the event that the agent interacts with the environment and continuously learns to maximize the expected return after time t , its main objective is to attempt to accrue as many rewards as possible during that interaction. Formula illustrates the procedure by Eq. (1)

$$R_t = r_{t+1} + \gamma \cdot r_{t+2} + \gamma \cdot r_{t+3} + \dots = \sum_{k=0}^{\infty} \gamma^k \cdot r_{t+k+1} \quad (1)$$

By establishing a value function that is split into a state-value function and a state-action-value function, RL creates link between optimal criterion as well as ideal policy. A state-value function is anticipated return that an agent using

policy π in state s is projected to get. Here is how the equation is displayed by Eq. (2)

$$V^{\pi}(s) = E_{\pi} \{ R_t \mid s_t = s \} = E_{\pi} \left\{ \sum_{k=0}^{\infty} \gamma^k r_{t+k+1} \mid s_t = s \right\} \quad (2)$$

where expected value under strategy π is indicated by the symbol E_{π} . When the best course of action is chosen from among all of these potential state-value functions, an optimal state-value function exists. Following defines expressions for the optimal state-value function as well as optimal strategy by Eq. (3)

$$\pi^* = \operatorname{argmax}_{\pi} V^*(s) \quad V^*(s) = \max_z V^*(s) \quad (3)$$

Similarly, expected return of agent carrying out action a in state s defined as state-action-value function. This function's equation is written as follows by Eq. (4)

$$Q^z(s, a) = E_z \{ R_t \mid s_t = s, a_t = a \} = E_{\pi} \left\{ \sum_{i=0}^{\infty} \gamma^i r_{t+i+1} \mid s_t = s, a_t = a \right\} \quad (4)$$

Expression of optimal approach is as follows in order to obtain optimal Q-value function by Eq. (5)

$$\pi^* = \operatorname{argmax}_{\pi} Q^*(s, a) \quad (5)$$

Bellman optimization equation is utilized to solve optimal Q-value function in order to further express it. The following is the final expression by Eq. (6)

$$Q^*(s, a) = R + \gamma E_j [V^*(s')] = R + \gamma \sum_{s' \in S} P(s' \mid s, a) V^*(s') \quad (6)$$

The technique is a hybrid multilevel symmetric key and asymmetric key technique that encrypts data at client end before uploading it to an online cloud storage provider. When moving data between their own data centers, many service providers don't encrypt it, which raises risk of data loss, privacy violations, government intrusions, theft of intellectual property, eavesdropping. Therefore, in order to mitigate such dangers, we can encrypt data at client side prior to uploading it to the cloud storage provider. Data encryption before it is transferred offers an added degree of security, even if SSL (Secure Sockets degree) is used to keep data private since it creates an encrypted link between a web server and browser while data is in transit. Consequently,

the primary goal of this algorithm—to safeguard data while it is in transit—is achieved.

It is paired with an ECC based on asymmetric cryptography and symmetric cyphers to provide data encryption based on EC private key and decryption by the associated EC public key. Although installing and designing public-key methods can be difficult, they provide a very useful and safe means of sharing data. EC is a smaller key size that enables safe authentication, key creation, and exchange mechanisms, while also requiring less bandwidth when communicating keys over a network via cloud servers. The approach employs elliptic curve cryptography (ECC) for secure key exchange and symmetric encryption for data protection. Encryption is done at the client side, and decryption on the cloud. The encryption/decryption process is described, but adding key generation, distribution, and management enhances trust in the system's security. Integrated Encryption Scheme with Elliptic Curve (ECIES) Choose a curve was frequently employed when we needed to exchange keys and sign something using just our public key and private key, then verify it using our public key. However, we could also utilize this method to encrypt data. These are the foundational ideas of encryption using elliptic curves. In order to find our public key, which is an x-y point, we first produce a random number for our private key, which is gradient of line that will originate from an elliptic curve (G) point. On this curve, we have an X and a width, but a private key is merely a scalar describing user. Hence, the inquiry at hand concerns whether the public key, denoted as Q, is equivalent to the private key times G by Eq. (7)

$$Q = P \cdot g \quad (7)$$

Finding value of P in the oval when we already know values of Q and G is extremely challenging if you use the huge private prime Y large integer method. Equation for elliptic curve cryptography provided by Eq. (8)

$$y^2 = x^3 + ax + b \quad (8)$$

The curve and values in this equation are represented by the points x and y, which correspond to the curve's values, and the points that define the curve, represented by the values a and b. Thus, elliptic key encryption functions as follows. The Integrated Encryption Scheme (IES) operates differently from our key exchange method in that it generates a symmetric key from public key of recipient of the data, whereas our method generates a key on the order side. This is because we use certain standards that are considered safe. represent an ECIES, which is a public-key method that offers symmetric encryption keys, digital signatures, and authenticated encryption. It is paired with an ECC based on

asymmetric cryptography and symmetric cyphers to provide data encryption based on EC private key and decryption by the associated EC public key. Public-key methods are challenging to install and design, but they offer an extremely helpful and secure way to share data. EC is a lower key size that allows for less bandwidth consumption for transmitting keys over a network via cloud servers, as well as safe authentication, key creation, and exchange techniques. Elliptic Curve Integrated Encryption Scheme (ECIES) chooses a curve that is frequently used when we want to exchange keys and sign documents using just our public key and private key, followed by a public key verification. Also use this technique to encrypt data.

i. Data optimization using Fuzzy Logic based - Firefly Whale Meta-heuristic Binary Swarm Algorithm (FL-FWMBSA).

Fuzzy logic is used in our trust management system. The partitioned database is represented by Let I, which stands for training data set TR I and testing data set TE I. Fuzzy system and its resulting fuzzy rules are designed using the training data. When assessing the classification's performance, the testing data set is taken into account. Fuzzy rule as well as fuzzy membership function are used to analyse the fuzzy system. The triangle function—s, t, and u in y(z) fuzzy set—contains three vertices, which are used to evaluate fuzzy membership function. In order to determine membership value, utilise Eq. (9).

$$y(z) = \begin{cases} 0 & \text{if } v \leq s \\ \frac{v-s}{t-s} & \text{if } s \leq v \leq t \\ \frac{u-v}{u-t} & \text{if } t \leq v \leq u \\ 0 & \text{if } v \geq u \end{cases} \quad (9)$$

Defining the membership function: Number of membership functions in fuzzy classifier is calculated using discretization function of each attribute. For every membership function, there are defined values for vertices s, t, u. To calculate the value for t, the highest value of u and the minimum value of s are chosen for each interval. The fuzzy system's categorization process makes use of both fuzzy rules and fuzzy membership function.

- i) Rule base in fuzzy method: Fuzzy rule set, which is sorted utilizing fuzzy rule based on GA, is produced as $J = \{J_p; 1 \leq p \leq q-K\}$. The fuzzy rule is present in rule base as L1 L2 L3 and L4 correspondingly representing low, medium, extremely low, very low decision.
- ii) Membership function of fuzzy: Additionally created is membership function for each attribute as well as its corresponding value for triangle membership. The test

data is changed into a fuzzified value in the testing data set. Linguistic value is then further transformed into a fuzzy score by matching input of fuzzified with fuzzy rules. Fuzzy score is utilized to generate optimal value, output produced by the fuzzy classifier is indicated by the notation h_j .

To support IDS, the main goal of improved FA is to create an autonomous as well as efficient network monitoring function. Effective network monitoring is beneficial for incident response and troubleshooting. Moreover, the suggested algorithm alerts the intrusion detection system to closely monitor dubious nodes, since these nodes can necessitate load balancing or intrusion investigation. In any scenario, it will enhance firefly network's overall performance and offer more assistance to the IDS.

A stochastic global optimisation technique inspired by nature is the firefly algorithm. The fundamental concept of this swarm-based meta-heuristic method is to generate an objective function for given issue based on luminosity of every firefly. This function instructs swarm of fireflies to move toward brighter fireflies in pursuit of optimal solutions. Every time an iteration is performed, the location of firefly i is determined as Eq. (10)

$$X_i^{t+1} = X_i^t + \beta^{-\gamma r_{ij}^2} (X_j^t - X_i^t) + \alpha_t \varepsilon_t, \quad (10)$$

where ε_t is vector taken from a Gaussian distribution, α_t is step size, r is distance between nodes x_i and x_j , β is attractiveness, η is absorption coefficient. In this case, β indicates the firefly's brightness, which is computed as Eq. (11)

$$\beta = \beta_0 e^{-\gamma r^2} \quad (11)$$

In the proposed approach, control is applied to $\beta_{\min}$. Accordingly, every node is first assigned a minimum brightness $\beta_{\min}$ and then has its brightness modified based on an initial brightness β_{init} by Eq. (12)

$$\beta_{\text{new}} = (\beta_{\text{init}} - \beta_{\min}) e^{-\gamma r^2} + \beta_{\min}. \quad (12)$$

In this case, Eq. (12) is changed to Eq. (13) in order to guarantee that the nodes' brightness stays above the set minimum value $\beta_{\min}$. The brightness is primarily determined by the Euclidean distance, or the distance between two firefly, and the absorption coefficient, or γ .

$$r = d(p, q) = \sqrt{\sum_{i=1}^n (q_i - p_i)^2} \quad (13)$$

Suggested technique effectively reaches out to several nodes while avoiding problems associated with local maxima by

utilising advantages of swarm optimisation and evolutionary computation. By being independent and able to roam throughout the network, each firefly in the suggested method avoids local minimum circumstances. The likelihood of each firefly calculating the network health in the same way is extremely low. Furthermore, the primary goal of the suggested approach is to confidently provide the IDS system with an early alert. WOA is a meta-heuristic method that draws inspiration from nature and emulates social behaviour of humpback whales. Bubble-net hunting tactic served as the model for the algorithm.

We will use following notation: $X_i = x_1^i, x_2^i, \dots, x_M^i$ is a vector that contains M decision variables for every candidate solution i in population. Every decision variable j of i -th solution (x_j^i) has a value that falls inside the range $[lb_j, ub_j]$, where lb_j and ub_j stand for search space's lower and upper bounds in j -th dimension. The fitness of the i -th solution is indicated by the notation $f_i(\sim X_i)$, and the value of the objective function for the same solution is indicated by the notation $f(\sim X_i)$. We included an extra variable trial_i for every solution $\sim X_i$ in the population. When a solution $\sim X_i$ cannot be improved upon in each iteration, its trial_i value is increased by one. A specific solution is removed from population and replaced with a new, randomly generated solution within search space by using following expression when trial value of that solution hits a predefined threshold value known as the limit by Eq. (14)

$$x_i^j = \theta \cdot (ub_j - lb_j) + lb_j \quad (14)$$

where θ is a uniformly distributed number in interval $[0, 1]$, x_j^i is j -th decision variable of i -th solution. This exploration technique maintains population diversity and speeds up convergence considerably, especially in early stages of method execution. Furthermore, such a technique can effectively steer search process towards other search domain regions when it becomes stuck in some of the suboptimal locations.

Optimization issues involving discrete functions, such as scheduling, routing, and integer programming, handled with the help of metaheuristics. Binary particle swarm optimization (BPSO) is proposed, which looks for patterns in the binary search space. Aiming to strike a balance between extraction and exploration, BPSO combines global and local search algorithms. The optimum location for each particle ($p\text{Best}$), the largest location conceivable worldwide ($p\text{Global}$), and the particle's position are used to continually update the velocities of all particles ($g\text{Best}$). BPSO employs only the particle's current velocity, as opposed to PSO, which modifies each particle's position based on its current position and velocity. A common method for updating particle positions is to use the sigmoid function. Furthermore,

one can use binary values to display the direction and velocity of particle migration within a population. Every particle in swarm represents a possible solution that could eventually result in a practical one. Put another way, this implies that the optimal solution may lie in subatomic particle k . As a result, each particle in this process goes through an equal number of iterations until one that consistently produces desired optimal response is found. This method is based on frequent comparisons between these particles. Repetition of every particle same number of times (5), as indicated by the equation, is one way to accelerate the process. Because it is a meta-heuristic algorithm, BPSO is less useful in practical situations.

Task scheduling as well as load balancing are carried out with the use of an appropriate BPSO algorithm architecture. There are three parts to this structure. The user requirements are divided into distinct jobs in the first module. The cloud administrator is in charge of this duty. Three modules rely on many hosts, and the virtual machine manager keeps track of every virtual machine that is being utilized to perform user-initiated operations. This is a scalable approach when applied correctly. Each particle's position and velocity are examined over a number of rounds of t to produce a fitness solution. The quantity of tasks, particles, virtual machines (VMs), iterations determines the time complexity of this phase. You have to use these two extremes to battle in this situation. Think about how many times it is repeated. We need to figure out how many iterations and particles we skipped in our search in order to find the perfect answer. Since all iterations and particles have ended, temporal complexity equals zero (spnm).

ii. Combined Cryptography and Optimization Process.

The system integrates FARCA (Federated Adversarial Reinforcement Cryptography Algorithm) for secure data migration, employing federated learning to locally train models without data sharing and adjustable encryption via reinforcement learning. The Firefly Whale Meta-Heuristic maximizes the process by searching for solutions with the firefly algorithm and improving them with the whale optimization algorithm. The algorithms collectively provide cloud data migration that is both secure and high-performance.

Pseudo-code for Combined Cryptography and Optimization Process:

1. Input `data_to_migrate`.
2. Encrypt data using FARCA.


```
output encrypted_data=FARCA.encrypt(data_to_migrate).
if encryption failed: error("Encryption failed") and return.
```
3. Optimize data using Firefly Whale Meta-Heuristic

- ```
output optimized_data=FF-WHA.optimize(encrypted_data).
if optimization failed: error("Optimization failed") and return.
```
4. Transfer data to cloud
 

```
output cloud_data=transfer_data_to_cloud(optimized_data).
if transfer failed: error("Data transfer failed") and return.
```
  5. Decrypt data on cloud
 

```
output decrypted_data = FARCA.decrypt(cloud_data)
if decryption failed: error("Decryption failed") and return
```
  6. Evaluate performance: if metrics meet threshold, return "Success", else error("Metrics below threshold")

## 6 Results and discussion

**Simulation setup-** We are only interested in the evolving stream mining random methods available in this software environment, despite the fact that there are many static ones as well. Random algorithms that are always evolving are concerned with changes in idea or distribution within the data stream. Every experiment is conducted on a single PC running Microsoft Windows 7 Professional 64-bit, equipped with an Intel Core i7-2600 3.4 GHz processor, 8.00 GB DDR3 main memory, and a 1.0 TB SATA hard disk drive.

**Dataset description: CIC-IDS2017-** The Bprofile method was employed in the CICIDS dataset, which was produced in partnership with University of New Brunswick and the Canadian Institute of Cybersecurity. In order to infer abstract behaviors for 25 users, network traffic that was generated made use of the file transfer protocol (FTP), secure shell protocol (SSH), email protocol, and HTTP and [HTTPS.CI-CIDS-2017](https://ci-cids-2017) dataset was utilized since it has several attributes that can be used to identify abnormalities, as well as more complex features and a large amount of traffic. Between hostile and benign traffic, entropy was 0.716, across attack kinds, it was 0.523. The balance between the sorts of attacks was relatively lower when compared to malicious vs. benign traffic.

**UNSW-NB15** The IXIA PerfectStorm program was utilized at lab of Australian Centre for Cyber Security (ACCS) to generate the UNSW NB15 dataset. CSV (comma-separated values) source files are part of the dataset. The dataset was created using an IXIA traffic generator connected to three servers via transmission control protocol (TCP). Three clients and two TCP dump servers were connected by a router.

As a result, the TCP dump produced CSV files. However, there was an issue with UNSW NB15 dataset's data's veracity.

This is because many attacks are classified as “generic,” yet it's not always clear what kind of attack they are. When comparing the anomalous and normal types, entropy was 0.548. Attack types alone have an entropy of 0.514, which suggests that they are generally somewhat more unbalanced than all other traffic categories combined.

**DS20S** This dataset also contains the communication between the various IoT nodes. All of these nodes are connected via distributed smart space orchestration system (DS20S), which is a common middleware. T=user-service communication in this instance was captured and stored as a CSV file. The dataset comprised 357,952 samples and 13 characteristics, whereas the IoT-based system contained window, light, and temperature controllers.

**KDD Cup 1999** The KDD Cup 1999 dataset is one of the most widely used datasets for detecting network intrusion. DARPA Intrusion Detection Evaluation Program was in use in this form back in 1998. Massachusetts Institute of Technology (MIT) Lincoln Labs collected raw TCP data in packet traces from an air force local area network (LAN) for approximately nine weeks. DoS attacks include R2L remote machine compromises and syn-floods. This was a huge dataset with about forty-one features. Important TCP properties were also accessible, such the source and destination bytes. Nevertheless, there was a duplication of data between the testing and training sets, and it was discovered that some crucial components—like IP addresses—were absent. Furthermore, the data were rejected since they came from synthetic generation and were obtained more than 20 years ago, making them outdated.

**Security analysis in cloud data migration:** The length of each chromosome is 16 characters encoded to 128 bits, and the initial population size is estimated to be 200. Using a loop, each person is delivered to the fitness function individually. Since the fitness function is a maxima function, the person with the highest fitness value will be chosen for additional consideration. Following this procedure, choose two people and execute byte-wise one-point crossover, with a random number being used to determine the crossover point. Once the crossover is complete, obtain the chosen

people's children. Next, the input for the mutation process is the result of the preceding step. The final key, which is required for the encryption procedure, is obtained following mutation.

## 7 Proposed model analysis

Table 1 shows parametric analysis of proposed technique for various cloud network security analysis dataset. Dataset analysed are CIC-IDS2017, UNSW-NB15, DS20S, KDD Cup 1999 dataset in terms of throughput, PDR, latency, QoS, data integrity.

The above Fig. 3 (a)-(c) shows parametric analysis of various cloud network security analysis dataset. The selected datasets—CIC-IDS2017, KDD Cup 1999, UNSW-NB15, and DS20S—are suitable for the testing of cloud data migration security. They contain protocol behaviors and traffic patterns that define cloud environments. The above features allow for the analysis of encryption, latency, throughput, and data integrity for the testing of the proposed model. For CIC-IDS2017 dataset the proposed model attained throughput 85%, PDR 84%, latency 82%, QoS 80%, data integrity 86%. throughput 92%, PDR 89%, latency 87%, QoS 85%, data integrity 92% for UNSW-NB15; for DS20S dataset proposed model attained throughput 95%, packet delivery ratio 93%, latency 94%, QoS 92%, data integrity 96%; proposed model throughput 98%, PDR 96%, latency 97%, QoS 95%, data integrity 98% for KDD Cup 1999 dataset.

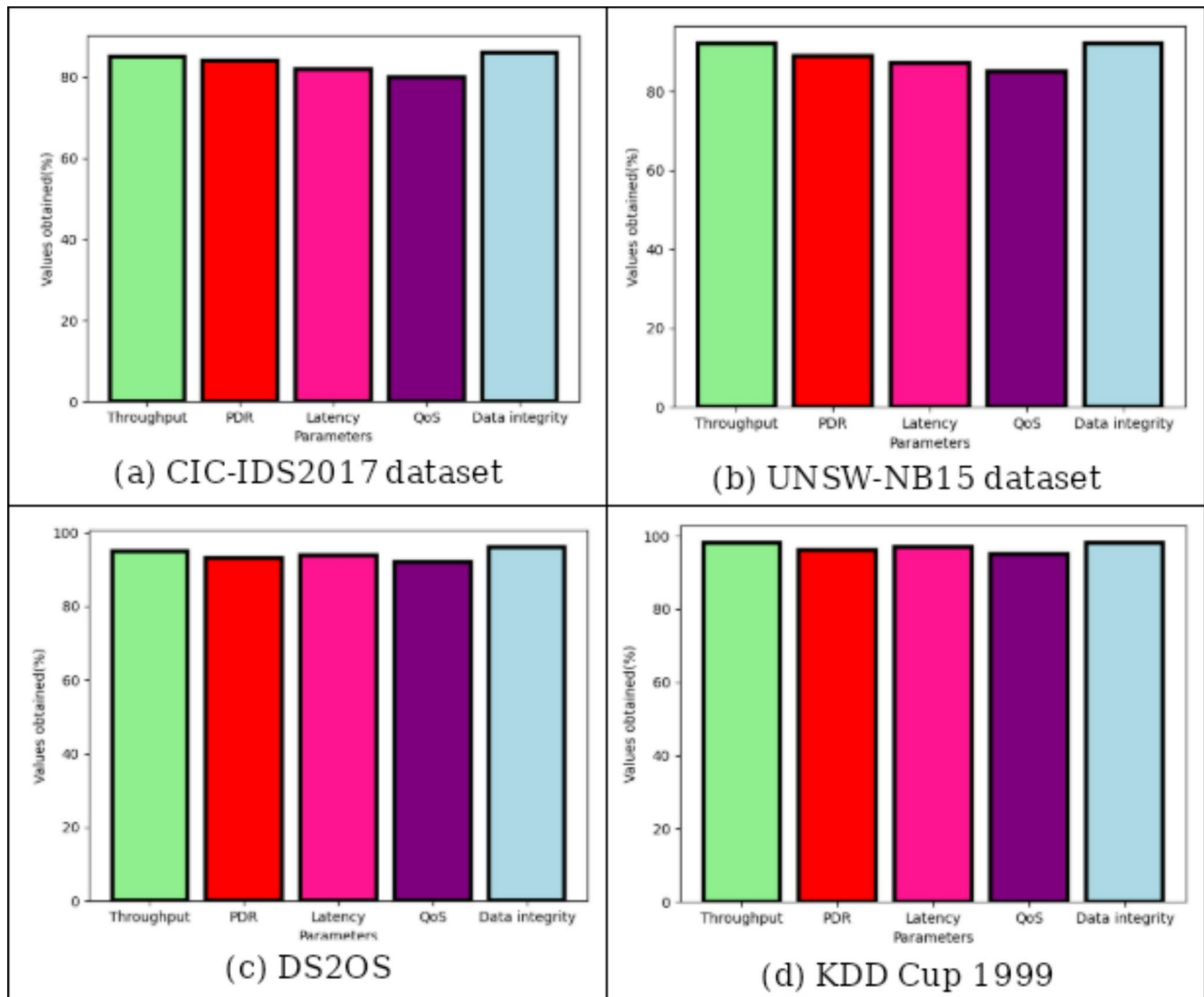
## 8 Comparative analysis

Table 2 shows parametric analysis for Comparative analysis for various cyber security dataset based on QKD. Here the dataset analysed are CI-CIDS-2017 [4], UNSW NB15 [5], DS20S [6], KDD CUP 1999 [7] in terms of PDR, Data integrity, Throughput, LATENCY, QOS.

The above Fig. 4 (a)-(e) shows comparative analysis for Data integrity, Throughput, LATENCY, QOS. Proposed technique Packet delivery ratio of 84%, existing PBE-ENCR 75% and SVM-GA 79% of PDR for CI-CIDS-2017 dataset; for UNSW NB15 dataset Proposed technique PDR of 89%, existing PBE-ENCR 80% and SVM-GA 85% of PDR; Proposed technique PDR 93%, existing PBE-ENCR 76% and SVM-GA 84% of PDR for DS20S dataset; for KDD CUP 1999 dataset Proposed technique PDR of 96%, existing

**Table 1** Parametric analysis of proposed technique for various cloud network security analysis dataset

| Dataset             | Throughput | Packet delivery ratio (PDR) | Latency | QoS | Data integrity |
|---------------------|------------|-----------------------------|---------|-----|----------------|
| <b>CIC-IDS2017</b>  | 85         | 84                          | 82      | 80  | 86             |
| <b>UNSW-NB15</b>    | 92         | 89                          | 87      | 85  | 92             |
| <b>DS20S</b>        | 95         | 93                          | 94      | 92  | 96             |
| <b>KDD Cup 1999</b> | 98         | 96                          | 97      | 95  | 98             |

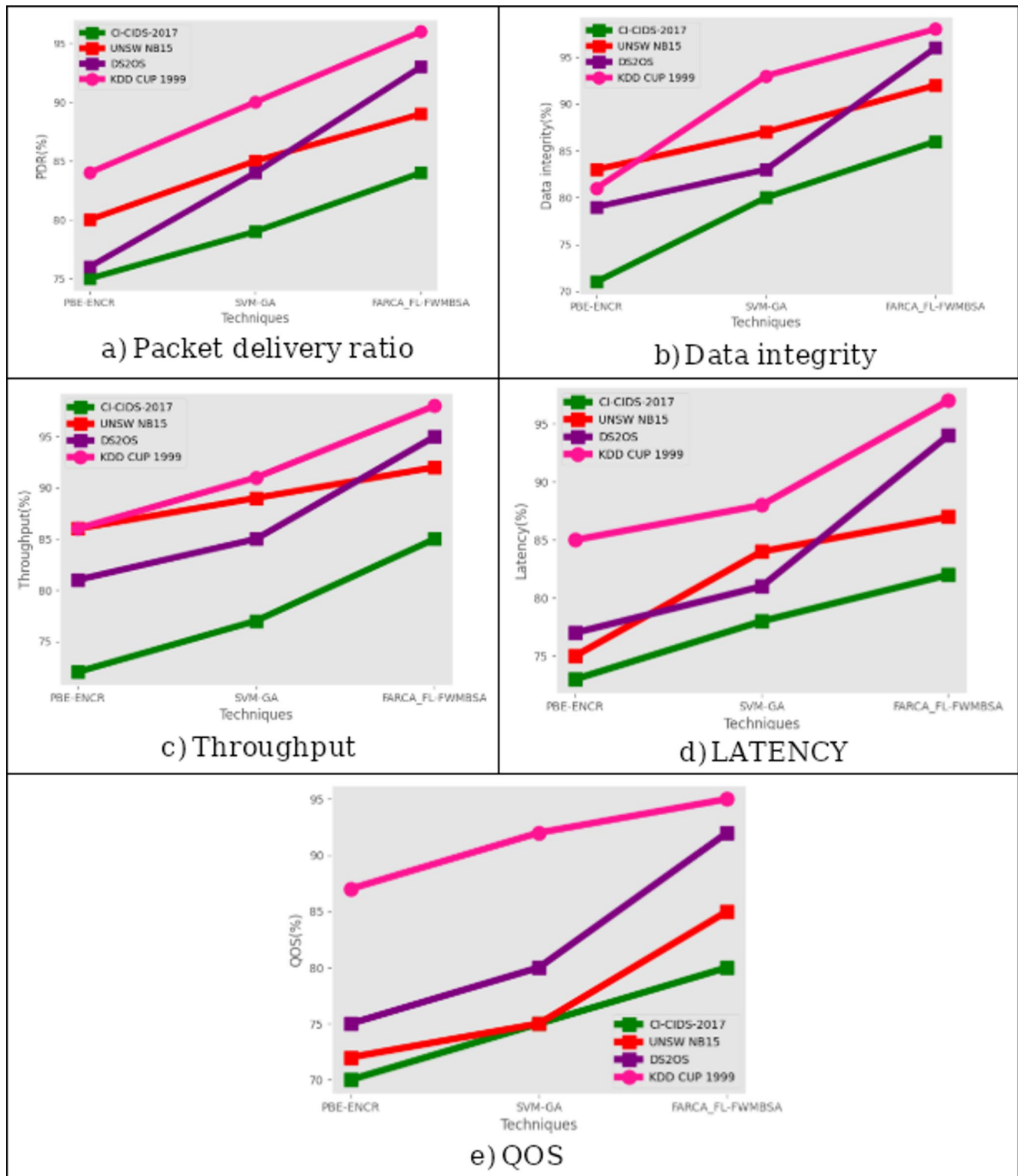


**Fig. 3** (a)-(d) Parametric analysis of proposed model for (a) CIC-IDS2017, (b) UNSW-NB15, (c) DS2OS, (d) KDD Cup 1999

**Table 2** Comparative based on various cloud network security analysis dataset

| Dataset             | Techniques      | Packet delivery ratio (PDR) | Data integrity | Throughput | Latency | QoS |
|---------------------|-----------------|-----------------------------|----------------|------------|---------|-----|
| CI-CIDS-2017<br>[4] | PBE-ENCR        | 75                          | 71             | 72         | 73      | 70  |
|                     | SVM-GA          | 79                          | 80             | 77         | 78      | 75  |
|                     | FARCA_FL-FWMBSA | 84                          | 86             | 85         | 82      | 80  |
| UNSW NB15<br>[5]    | PBE-ENCR        | 80                          | 83             | 86         | 75      | 72  |
|                     | SVM-GA          | 85                          | 87             | 89         | 84      | 75  |
|                     | FARCA_FL-FWMBSA | 89                          | 92             | 92         | 87      | 85  |
| DS2OS<br>[6]        | PBE-ENCR        | 76                          | 79             | 81         | 77      | 75  |
|                     | SVM-GA          | 84                          | 83             | 85         | 81      | 80  |
|                     | FARCA_FL-FWMBSA | 93                          | 96             | 95         | 94      | 92  |
| KDD CUP 1999<br>[7] | PBE-ENCR        | 84                          | 81             | 86         | 85      | 87  |
|                     | SVM-GA          | 90                          | 93             | 91         | 88      | 92  |
|                     | FARCA_FL-FWMBSA | 96                          | 98             | 98         | 97      | 95  |





**Fig. 4** Comparative analysis for various remote sensed soil image dataset

PBE-ENCR attained 84% and SVM-GA 90% of PDR. For CI-CIDS-2017 dataset, proposed method 86% data integrity, PBE-ENCR 71%, 3D CAD achieved 80% data integrity. For UNSW NB15 dataset, proposed method 92% data

integrity, PBE-ENCR 83%, 3D CAD achieved 87% data integrity. For DS2OS dataset the proposed technique a data integrity of 96%, whereas existing PBE-ENCR achieved 79% and SVM-GA 83% of data integrity; for KDD CUP

1999 dataset, proposed technique a data integrity of 98%, existing PBE-ENCR achieved 81%, SVM-GA achieved 83% of data integrity. Proposed method an Throughput of 85% for UNSW NB15 dataset, existing PBE-ENCR attained 72% and SVM-GA attained 77%. For the KDD CUP 1999 dataset, proposed technique an Throughput of 98%, existing PBE-ENCR attained 86%, SVM-GA 91%. For the DS2OS dataset, proposed technique achieved an Throughput of 95%, existing PBE-ENCR 81%, SVM-GA 85% of the Throughput. Throughput of suggested method was 85%, PBE-ENCR was 72%, SVM-GA was 77% for CI-CIDS-2017 dataset. proposed approach achieved an LATENCY of 82%, while the previous PBE-ENCR achieved 73% and SVM-GA achieved 78% of the LATENCY for CI-CIDS-2017. The proposed method achieved an LATENCY of 87%, while the current PBE-ENCR 75% and SVM-GA 84% of LATENCY for UNSW NB15 dataset; for the DS2OS dataset, the proposed method achieved an LATENCY of 94%, the current PBE-ENCR attained 77%, and SVM-GA attained 81% of the LATENCY; this was also the case with the KDD CUP 1999 dataset While the current PBE-ENCR achieved 85% and SVM-GA 89% of LATENCY, the proposed approach achieved 97%. For CI-CIDS-2017 dataset the proposed model attained QOS of 80%, current PBE-ENCR attained 70%, and SVM-GA attained 75% of QOS; The suggested approach produced an QOS of 85% for the CI-CIDS-2017 dataset, compared to 72% for the standard PBE-ENCR and 75% for SVM-GA. The outcomes for the UNSW NB15 dataset were comparable. For the DS2OS dataset, proposed technique obtained QOS of 92%, current PBE-ENCR attained 75%, and SVM-GA attained 80% of QOS; for the KDD CUP 1999 dataset, proposed technique QOS of 95%, current PBE-ENCR attained 87%, and SVM-GA attained 92% of QOS. Statistical significance, via tests like paired t-tests, determines that gains in such significant indicators as data integrity and packet delivery ratio are great ( $p < 0.05$ ). There are 95% confidence intervals that validate also such improved performance over existing methodologies.

## 9 Discussion

The proposed approach is better in latency from efficient optimization by the firefly whale swarm algorithm, which reduces delay times of transmission. Integrity in data is improved from the powerful federated adversarial cryptography employed to secure data without revealing raw data, rendering overall encryption more robust. We suggested a novel method of moving data between cloud servers by combining steganography as well as cryptography. Utilizing one's complement approach, we employ a relatively basic yet powerful technique in the cryptography process

to encrypt data. It employed the symmetric key method, in which encryption and decryption keys are shared by sender as well as recipient. Symmetric key method's strength is that it creates a dynamic value for the private key, which makes it extremely safe because no one can access the private key or even hold it; instead, it is updated for each piece of data that needs to be communicated. Data can be moved between public clouds and data centres, as well as between clouds and data centres. The actions of cloud intruders are curtailed by assigning temporary tickets with restricted privileges and a limited active period. Two tickets received in the system point to a potential attack. Reducing the privileges prevents the hacker from accessing the stolen information. As a result, the data integrity is maintained. For further pragmatic utility, upcoming work will calibrate the model using real-world cloud migration traces from hybrid deployments. This includes encryption delay, access controls, and transfer rates during actual service migrations. Such integration will test the performance of the model in real-world applications beyond test data.

## 10 Advantages and limitation

The suggested method offers a safe way to move data to cloud servers. It uses steganography and cryptographic techniques that give the client's sensitive data files many layers of protection. The symmetric strategy is used in cryptography, and it is less expensive than the asymmetric approach. To make matters better, it always uses dynamic numbers for calculations. data transfer security because it creates a security channel between source as well as destination, allowing the transfer of encryption keys, random keys, and message authentication codes. Additionally, the least privilege migration ticket permits the best possible data transfer. Because the ticket is only intended for one use, it is simple to identify intruders when the ticket is used more than once. Finally, the employment of a shared key and random key improves PBE-based encryption, increasing the degree of data protection. Blocks of the data can be kept separately. For transmission, each individual data block might be encrypted independently. One drawback of the suggested model is that it requires less time to encrypt and transport data. Moreover, data blocks are divided into smaller pieces to increase transmission speed as well as security. Cloud storage solutions can all use this foundation.

## 11 Conclusion

Transferring data, other business necessities into a cloud computing environment is known as cloud migration. Using machine learning and cryptography approaches, the suggested model in this research proposes a novel method for secure data movement in cloud networks. Here, data migration is accomplished in a secure cloud environment through the use of the fuzzy logic-based firefly whale meta-heuristic binary swarm method for data optimisation and the federated adversarial reinforcement cryptography algorithm. The suggested system includes user registration, authentication, data encryption, data storage, activity tracking for users, controller data mining. These strategies can all be used for real-time applications and are executed with perfection. Since we implemented a distributed architecture, our system's security has increased. The suggested technique has proven to have a strong security level as well as noticeable improvement in encryption/decryption, offering high security as well as cheap computing cost, according to experimental results. Because of its quick data collecting and processing times, it is even more useful in field of cloud computing. Suggested algorithm put into hardware in the future, which could lead to considerably better outcomes.

**Author contributions** Kannan Srinivasan and Guman Singh Chauhan wrote the main manuscript text. Rahul Jadon and Rajababu Budda conducted the experiments and performed the data analysis. Venkata Surya Teja Gollapalli prepared Figs. 1, 2 and 3 and contributed to the methodology. Joseph Bamidele Awotunde participated in the literature review and assisted in refining the discussion and conclusion sections. All authors reviewed and approved the final manuscript.

**Data availability** No datasets were generated or analysed during the current study.

## Declarations

**Conflict of interest** The authors declare no competing interests.

**Patient consent statement** We have not harmed any human person with our research data collection, which was gathered from an already published article.

**Clinical trial registration** We have not harmed any human person with our research data collection, which was gathered from an already published article.

## References

- Thabit, F., Can, O., Wani, R.U.Z., Qasem, M.A., Thorat, S.B., Alkhazami, H.A.: Data security techniques in cloud computing based on machine learning algorithms and cryptographic algorithms: Lightweight algorithms and genetics algorithms. *Concurrency Computation: Pract. Experience*. **35**(21), e7691 (2023)
- Thamil Selvi, C.P., Lakshmana Kumar, R., Punitha, P.: Enhancing data security in the cloud using MECC-SIDH enhanced CL-HPAEKS scheme. *Wireless Netw.* **31**, 2477–2494 (2025)
- Punitha, P., Lakshmana Kumar, R., Revathi, S., Premalatha, R., Aiswarya, R.S.: Secured framework with a hash Function-Enabled keyword search in cloud storage services. *Int. J. Coop. Inform. Syst.* **33**(03), 2450001 (2024)
- Kaleem, M., Mushtaq, M.A., Jamil, U., Ramay, S.A., Khan, T.A., Patel, S., Hussain, S.K.: New efficient cryptographic techniques for cloud computing security. *Migration Lett.* **21**(S11), 13–28 (2024)
- Arunkumar, M., Ashok Kumar, K.: Malicious attack detection approach in cloud computing using machine learning techniques. *Soft. Comput.* **26**(23), 13097–13107 (2022)
- Almuzaini, K.K., Sinhal, A.K., Ranjan, R., Goel, V., Shrivastava, R., Halifa, A.: Key aggregation cryptosystem and double encryption method for Cloud-Based intelligent machine learning Techniques-Based health monitoring systems. *Comput. Intell. Neurosci.* **2022**(1), 3767912 (2022)
- Gupta, I., Singh, A.K., Lee, C.N., Buyya, R.: Secure data storage and sharing techniques for data protection in cloud environments: A systematic review, analysis, and future directions. *IEEE Access*. **10**, 71247–71277 (2022)
- Aruna, M.G., Hasan, M.K., Islam, S., Mohan, K.G., Sharan, P., Hassan, R.: Cloud to cloud data migration using self sovereign identity for 5G and beyond. *Cluster Comput.* **25**(4), 2317–2331 (2022)
- Tamizhselvan, C., Vijayalakshmi, V.: Cloud data access governance and data security using distributed infrastructure with hybrid machine learning architectures. *Wireless Netw.* **30**(4), 1–16 (2024)
- Ahmad, S., Mehruz, S., Urooj, S., Alsubaie, N.: Machine learning-based intelligent security framework for secure cloud key management. *Cluster Comput.* **27**(5), 1–27 (2024)
- Suganya, M., Sasipraba, T.: Stochastic gradient descent long short-term memory based secure encryption algorithm for cloud data storage and retrieval in cloud computing environment. *J. Cloud Comput.* **12**(1), 74 (2023)
- Kaur, H., Gargrish, S.: Evaluation of Secure Methods for Migrating Virtual Machines to the Cloud. In *2024 IEEE International Conference on Computing, Power and Communication Technologies (IC2PCT)* (Vol. 5, pp. 1961–1968). IEEE. (2024), February
- Mohammed, S., Nanthini, S., Krishna, N.B., Srinivas, I.V., Rajagopal, M., Kumar, M.A.: A new lightweight data security system for data security in the cloud computing. *Measurement: Sens.* **29**, 100856 (2023)
- Kaur, H., Anand, A.: Review and analysis of secure energy efficient resource optimization approaches for virtual machine migration in cloud computing. *Measurement: Sens.* **24**, 100504 (2022)
- Qamar, S., Amaan, M., Rahman, M.I., Aqeel, I., Shuaib, M., Khormi, I.M., Alam, S.: Cloud data transmission based on security and improved routing through hybrid machine learning techniques. *Soft. Comput.* (2023). <https://doi.org/10.1007/s00500-023-08417-0>
- Cherbal, S., Zier, A., Hebal, S., Louail, L., Annane, B.: Security in internet of things: A review on approaches based on blockchain, machine learning, cryptography, and quantum computing. *J. Supercomputing*. **80**(3), 3738–3816 (2024)
- Mangalagowri, R., Venkataraman, R.: Ensure secured data transmission during virtual machine migration over cloud computing environment. *Int. J. Syst. Assur. Eng. Manage.* (2023). <https://doi.org/10.1007/s13198-022-01834-8>
- Pon, P., V, K.: Blockchain based cloud service security architecture with distributed machine learning for smart device traffic

- record transaction. *Concurrency Computation: Pract. Experience*, **34**(3), e683. (2022)
19. Ganesan, S., Musam, V.S., Musham, N.K., Purandhar, N.: Scientific computing in the cloud: Impact of ant colony optimization, gradient descent, and bayesian decision models. *Int. J. Eng. Sci. Res.* **12**(4), 102–118 (2022)
  20. Altahat, M.A., Daradkeh, T., Agarwal, A.: Optimized encryption-integrated strategy for containers scheduling and secure migration in multi-cloud data centers. *IEEE Access.* **12**, 51330–51345 (2024)
  21. Kulkarni, S., Tripathi, R.K., Joshi, M.: A study on data security in cloud computing: Traditional cryptography to the quantum age cryptography. In: *System Design Using the Internet of Things with Deep Learning Applications*, pp. 147–174. Apple Academic (2023)
  22. Chauhan, G.S., Jadon, R., Budda, R., Gollapalli, V.S.T., Srinivasan, K., Prema, R.: Federated learning-based privacy-preserving deep learning for secure data analytics using differential privacy. *Int. Res. J. Adv. Eng. Sci.* **10**(2), 14–18 (2025)
  23. Bhavya, R., Kumar, M.G.V., Ramya, U.M., Janagi, R., Ganesan, M., Deepa, S.: Recognition of Secure Data Transmission in Cloud Platform using Deep Learning. In *2024 3rd International Conference on Applied Artificial Intelligence and Computing (ICAAIC)* (pp. 754–759). IEEE. (2024), June
  24. Poovendran, A.: A systematic literature review of the Elliptic Curve Cryptography (ECC) Algorithm for encrypting data sharing in cloud computing, *International Journal of Engineering & Science Research, IJESR/April. 2024/ Vol-14/Issue-2/1717–1736, ISSN 2277–2685.* (2024)
  25. Kumar, R.L., Subramanian, R., Karthik, S.: Hash function based keyword searchable encryption framework in cloud server using MD5 and MECC. *J. Inform. Hiding Priv. Prot.* **3**(4), 201–226 (2021)

**Publisher's note** Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.